Blu: The Macaw

Project Design Document

1. Design Intent

What experience are you trying to create?
The project aims to create an **interactive animatronic bird** that feels *aware* rather than reactive. Instead of instant responses, Blu pauses, observes, and responds subtly, creating the feeling that it **chooses** when and how to engage.

The experience is meant to feel quiet, curious, and slightly unpredictable—closer to interacting with a living creature than a machine.

Who is it for?
The experience is designed for:

- Exhibition visitors
- First-time users with no technical background
- Viewers observing from a distance as well as direct interactors

If solving a problem, what problem does it solve?
Many interactive objects feel mechanical because they react instantly and predictably.
This project explores how **behavior, timing, and restraint** can make objects feel more alive, even with minimal hardware.

2. Inspiration & References

https://www.behance.net/gallery/6052237/Blu-Jewel-Nigel-Rio

- ***How and when Blu moves is more important than how much it moves.***

3. User Journey

Step-by-step experience of the user:

ï **User starts…**
The user enters the space and notices a bird sitting quietly. It appears inactive, possibly observing.
ï **User interacts…**
As the user moves closer, Blu does not react immediately. After a short pause, it slowly turns its head, as if noticing the presence.
ï **System responds…**

- Blu tilts its head slightly, studying the user
- If the user stays, Blu engages with subtle movements
- If the user gets too close or touches suddenly, Blu reacts cautiously
- Gentle touch calms the bird; abrupt interaction startles it

4. Success Criteria

Define measurable success:

Functional (e.g., LED responds in <1 sec)

ï Proximity detection response triggered within 1–2 seconds

ï Servo movements are smooth and jitter-free
ï System runs continuously for extended periods without crashing
User experience (easy to use, intuitive)
ï Interaction feels intuitive without instructions

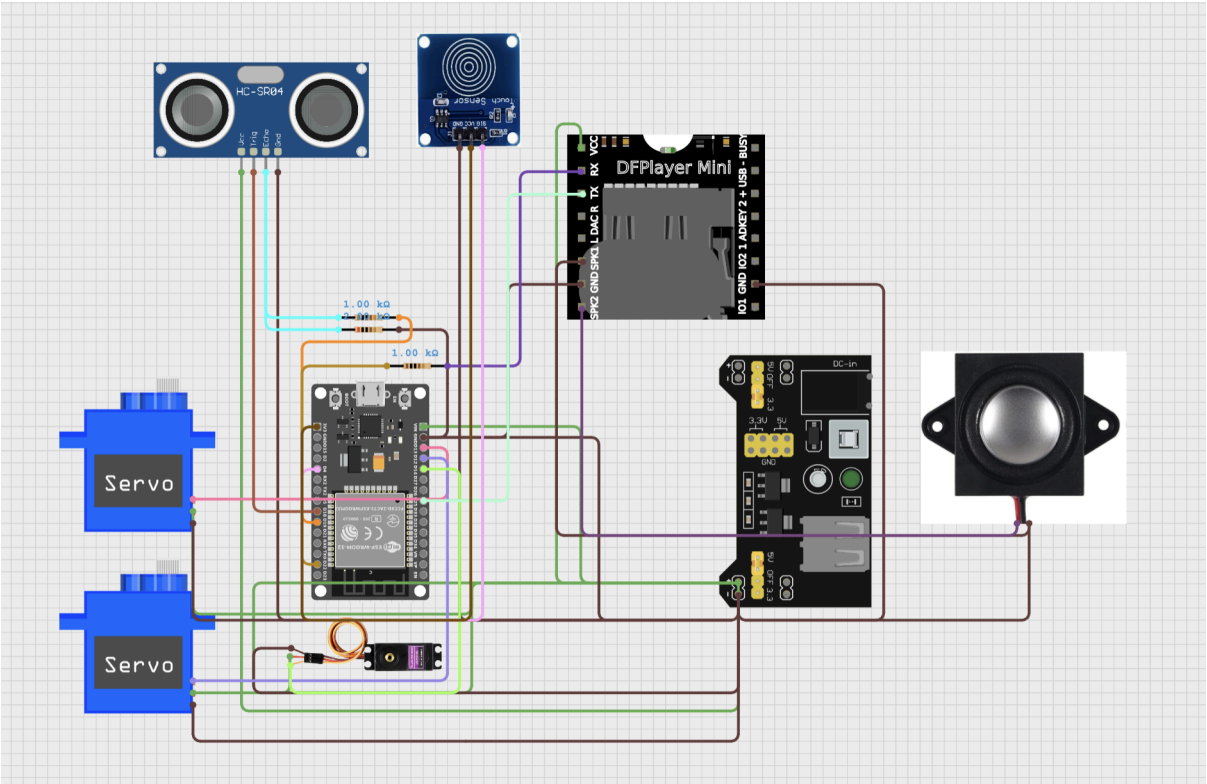
ï Users describe Blu as “curious,” “alive,” or “aware”
ï Movements feel intentional, not robotic
Reliability
ï Stable power delivery to servos

ï No random resets or sensor misfires
ï Consistent behavior across repeated interactions

5. System Overview

- Input sensors (proximity + touch)
- Microcontroller (ESP32)
- Animation software (Bottango)
- Servo motors (head movement)
- Physical bird structure

6. High-Level Block Diagram

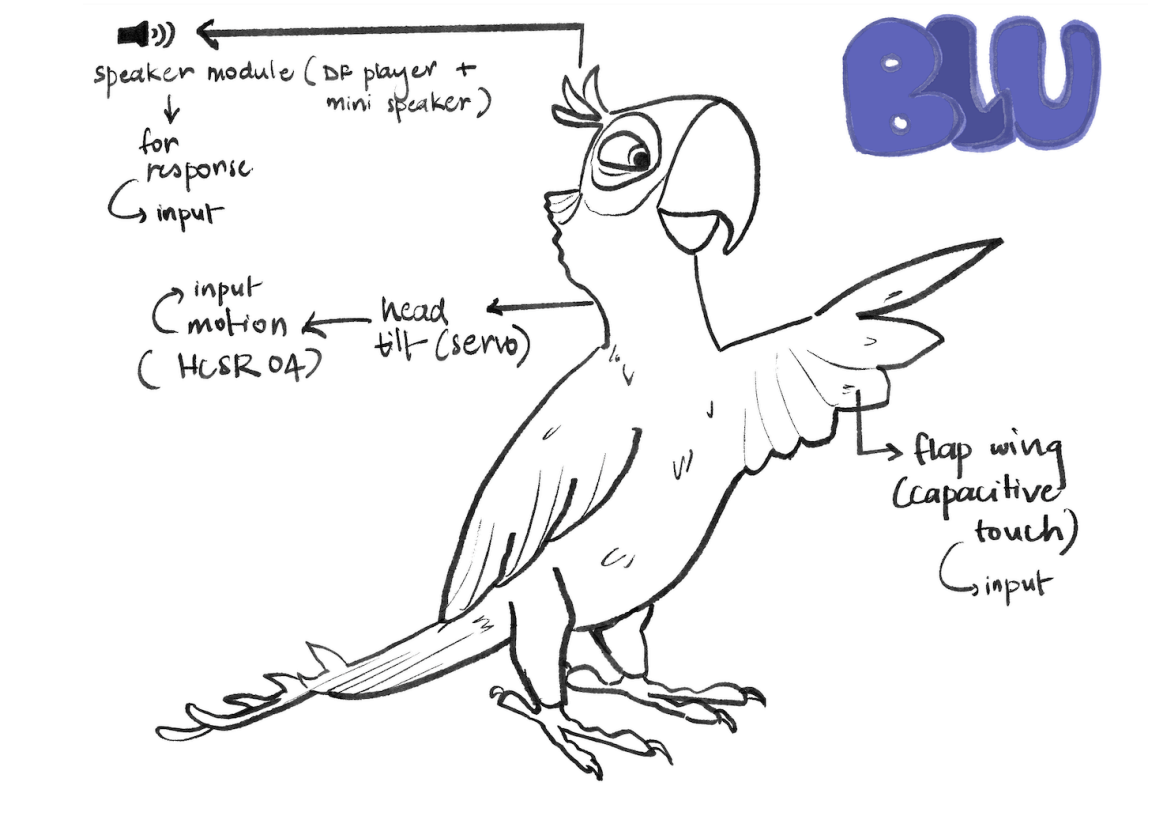


7. Input / Output Mapping

Input	Process	Output

User presence User distance change Gentle touch Sudden touch No input	Distance data interpreted State transition logic Calm trigger Startle trigger Idle loop	Head turns toward user Idle / Noticing / Engaging Slow head tilt Quick recoil motion Subtle breathing-like motion

8. Sketches & Concepts



9. Final System Architecture (Labelled)

- **Inputs**
  - Ultrasonic / IR proximity sensor
  - Capacitive touch sensor
- **Controller**
  - ESP32 microcontroller
  - Reads sensors and sends state triggers
- **Behavior & Animation**
  - Bottango software
  - State machine + animation timelines
- **Outputs**
  - Servo motor (head yaw – left/right)
  - Servo motor (head pitch – up/down)
  - Optional buzzer for minimal sound feedback
- **Physical Layer**
  - Lightweight bird body
  - Hidden wiring and mounted servos